

Prelecture Notes: In-Context Learning, Prompting & PEFT (Soft Prompting + LoRA)

1. Big Picture

Large language models can adapt to new tasks in two main ways: **In-Context Learning (ICL)**: The model learns from examples or instructions in the input. No weights are updated; learning happens temporarily during inference. **Fine-Tuning**: The model's parameters are updated with new data to permanently adapt to a task. This is powerful but computationally expensive.

Parameter-Efficient Fine-Tuning (PEFT) provides a middle ground: only a small subset of parameters are trained while keeping the base model frozen. This reduces compute, cost, and storage while maintaining strong task performance.

2. PEFT Methods Overview

Method	What's Trained	Where It Acts	Typical Size	Pros
Soft Prompting	Learnable input embeddings	Input layer	~0.1–1% of parameters	Simple, quick, no arch changes
LoRA	Low-rank matrices (A, B)	Inside attention/MLP layers	~0.5–2%	Stronger adaptation, works on many tasks

3. Key Intuitions

ICL = temporary learning from examples (no weight updates). **Soft Prompting** = train a few embedding vectors that steer the model through input context. **LoRA** = add small trainable matrices inside layers; model learns richer adaptations efficiently. LoRA formula: $W' = W + A * B^T$ — base weights stay frozen. Trade-off: Soft Prompting is fastest and smallest; LoRA adapts better for complex tasks.

4. Quick Prelecture Quiz

1. Why is in-context learning called “learning” even though no weights change?
2. In LoRA, what does “low-rank” mean and why is it efficient?
3. Which PEFT method would you choose if you want fastest adaptation and smallest memory use?

5. Lecture Focus Checklist

Understand how in-context learning differs from fine-tuning. Know the difference between **hard** (text) and **soft** (learned) prompts. Follow how LoRA inserts trainable low-rank adapters into transformer layers. Watch for examples showing PEFT's efficiency vs. full fine-tuning. Be ready for math snippets like $W' = W + A * B^T$ (LoRA update rule).